Math 550, S20 **Name:** Answer

Quiz 10

Instructions: **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (6 points) Given a helicoid parametrized by

$$\Phi(r,\theta) = (r\cos\theta,\, r\sin\theta,\, \theta), \quad 0 \le r \le 2,\, 0 \le \theta \le 2\pi.$$

(a) Find an equation for the plane tangent to this helicoid at the point $(-1, 1, 3\pi/4)$.

$$r\cos\theta = -1,\ r\sin\theta = 1,\ \theta = 3\pi/4 \Rightarrow r = 1/\sin(\tfrac{3\pi}{4}) = \frac{1}{\sqrt{2}/2} = \sqrt{2}$$

$$T_r(r,\theta) = (\cos\theta, \sin\theta, 0),\quad T_\theta(r,\theta) = (-r\sin\theta, r\cos\theta, 1)$$

$$\Rightarrow T_r(\sqrt{2}, \tfrac{3\pi}{4}) = (-\tfrac{\sqrt{2}}{2}, \tfrac{\sqrt{2}}{2}, 0),\quad T_\theta(\sqrt{2}, \tfrac{3\pi}{4}) = (-1,-1,1)$$

$$\Rightarrow N(\sqrt{2}, \tfrac{3\pi}{4}) = T_r(\sqrt{2}, \tfrac{3\pi}{4}) \times T_\theta(\sqrt{2}, \tfrac{3\pi}{4})$$

$$= \begin{vmatrix} i & j & k \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ -1 & -1 & 1 \end{vmatrix} = \frac{\sqrt{2}}{2} i + \frac{\sqrt{2}}{2} j + \sqrt{2} k$$

Thus the equation of tangent plane at $(-1, 1, \frac{3\pi}{4})$ is

$$\frac{\sqrt{2}}{2}(x+1) + \frac{\sqrt{2}}{2}(y-1) + \sqrt{2}(z - \tfrac{3\pi}{4}) = 0$$

$$x + y + 2z = \frac{3}{2}\pi$$

(b) Find the scalar surface integral $\iint_\Phi \sqrt{x^2+y^2}\, dS$.

$$N(r,\theta) = T_r(r,\theta) \times T_\theta(r,\theta)$$

$$= \begin{vmatrix} i & j & k \\ \cos\theta & \sin\theta & 0 \\ -r\sin\theta & r\cos\theta & 1 \end{vmatrix} = (\sin\theta) i + (\cos\theta) j + r k$$

$$\Rightarrow \|N(r,\theta)\| = \sqrt{\sin^2\theta + \cos^2\theta + r^2} = \sqrt{r^2+1}$$

Thus $\displaystyle\iint_\Phi \sqrt{x^2+y^2}\, dS = \iint_D \sqrt{x^2+y^2}\, \|N(r,\theta)\|\, dr\, d\theta$

$$\begin{matrix} 0 \le r \le 2 \\ 0 \le \theta \le 2\pi \end{matrix} \quad = \int_0^2 \int_0^{2\pi} r\sqrt{r^2+1}\, d\theta\, dr$$

$$= \int_0^2 2\pi r \sqrt{r^2+1}\, dr = \Big[\pi \tfrac{2}{3}(r^2+1)^{3/2}\Big]_{r=0}^{r=2}$$

$$= \frac{2\pi}{3}(5^{3/2} - 1)$$

Problem 2. (4 points) Let **S** be the portion of the surface $z = 1 - x^2 - y^2$ that lies above the xy-plane and suppose **S** is oriented up (i.e., the standard normal vector **N** faces up). Find the flux of the vector field $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ across **S**, i.e., $\iint_{\Phi} \mathbf{F} \cdot d\vec{\mathbf{S}}$.

$$\Phi(r,\theta) = (r\cos\theta, r\sin\theta, 1-r^2), \quad \underbrace{0 \le r \le 1, \quad 0 \le \theta \le 2\pi}_{D}$$

$$T_r(r,\theta) = \frac{\partial \Phi}{\partial r}(r,\theta) = (\cos\theta, \sin\theta, -2r)$$

$$T_\theta(r,\theta) = \frac{\partial \Phi}{\partial \theta}(r,\theta) = (-r\sin\theta, r\cos\theta, 0)$$

$$\Rightarrow N(r,\theta) = T_r(r,\theta) \times T_\theta(r,\theta)$$

$$= \begin{vmatrix} i & j & k \\ \cos\theta & \sin\theta & -2r \\ -r\sin\theta & r\cos\theta & 0 \end{vmatrix} = (2r^2\cos\theta)i + (2r^2\sin\theta)j + rk$$

face up

thus

$$\iint_{\Phi} F \cdot d\vec{S} = \iint_D (\overset{r\cos\theta}{x}i + \overset{r\sin\theta}{y}j + \underset{1-r^2}{z}k) \cdot (2r^2\cos\theta\, i + 2r^2\sin\theta\, j + rk)\, dr\, d\theta$$

$$= \int_0^1 \int_0^{2\pi} 2r^3\cos^2\theta + 2r^3\sin^2\theta + r(1-r^2)\, d\theta\, dr$$

$$= \int_0^1 \int_0^{2\pi} r^3 + r\, d\theta\, dr$$

$$= 2\pi \int_0^1 r^3 + r\, dr$$

$$= 2\pi \left[\frac{r^4}{4} + \frac{r^2}{2}\right]_{r=0}^{r=1}$$

$$= 2\pi \cdot \frac{3}{4} = \frac{3}{2}\pi$$